
PROJECT DESCRIPTION

ScanWise: AI-Powered Multi-Scan Medical Imaging and Report Interpretation System

Domain: MedTech / HealthTech / AI for Healthcare

Category: Software-Only, Cloud + Agentic AI + Computer Vision

Scope: Clinical Imaging Support, Patient Engagement, Early Triage Assistance

Platform: Firebase + Gemini + Google Cloud

Overview

Modern healthcare systems face increasing pressure due to population growth, shortage of radiologists, and high patient loads in government hospitals. The lack of timely interpretation of medical scans — especially in rural and semi-urban regions — leads to delayed diagnosis and poor health outcomes.

ScanWise is an intelligent cloud-based healthcare assistance platform that uses **multi-modal AI** to analyze different types of medical scans and lab reports, provide **high-level pattern interpretation**, and offer **triage-based urgency guidance** with **doctor-directed next steps**.

It enables patients and clinicians to upload:

- Chest X-rays
- Bone X-rays
- Brain MRI scans
- Ultrasound images
- Skin lesion photographs
- Scanned/PDF lab reports

and receive an **instant AI-generated summary** written in **simple, safe, non-diagnostic language**, helping them understand what might be visible in the scan and **how urgently a doctor should review it**.

Problem Statement

Many healthcare facilities — especially in Tier-2/3 India — lack access to trained radiologists 24×7. This results in:

- Delayed detection of critical conditions
- Poor patient awareness of scan findings
- Overcrowded tertiary hospitals due to lack of triage
- Unequal access to basic understanding of medical images

There is a need for a **software-only, scalable, low-cost** system that can:

1. Interpret multiple medical scan types safely
2. Provide patient-friendly explanations
3. Assist triage and clinical prioritization
4. Integrate seamlessly into existing hospital workflows

ScanWise directly addresses these issues using **Agentic AI** and **secure cloud healthcare architecture**.

Objectives

1. Automate **high-level visual pattern recognition** in multiple clinical imaging modalities.
 2. Assist with **urgency classification** to support triage.
 3. Provide **easy-to-understand summaries** for patients in India.
 4. Maintain safety by enforcing **no diagnosis, no prescription** rules.
 5. Enable **doctor involvement** via hospital locator and history viewer.
 6. Ensure security of medical data using **privacy-focused cloud design**.
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Key Features & Innovation

Feature	Description
Multi-scan Support	One model to handle X-ray, MRI, Ultrasound, Skin photos & lab reports
Gemini Agentic AI	Specialized agents collaborate for image + context reasoning
Cloud-based Medical Storage	Firestore + Storage with patient-controlled access
Triage-Based Urgency Scoring	EMERGENCY/PRIORITY/ROUTINE classification
Hospital Recommendation	Google Maps API for nearest hospital routing
Patient-Friendly Output	Results explained in simple English + disclaimers
Doctor Dashboard	Scan review & patient progression tracking
Scalable Healthcare Delivery	Works anywhere with a smartphone + internet

This makes ScanWise **unique** and **deployment-ready** for real clinical environments.

Technology Stack

- **AI Engine:** Gemini (Vision + LLM), Multi-agent framework
 - **Cloud Platform:** Firebase (Firestore, Storage, Auth, Functions, Hosting)
 - **Standards:** HIPAA-inspired access control, privacy-by-design
 - **APIs Integrated:**
 - Gemini Vision Pro
 - Google Maps/Places API
 - Firebase Security Rules
 - Optional: Cloud Healthcare API for DICOM support
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Safety, Privacy & Ethics

- No medical diagnosis or decision-making
- Strong disclaimers + clinical verification required
- Role-based access (patient/doctor)
- End-to-end secure data handling
- Red-flag escalation only to **human** medical professionals

Result: **Assistive**, not **replacing** clinicians.

Expected Outcomes

- Accelerated **scan interpretation** for high-volume healthcare settings
 - Improved **patient understanding** and engagement
 - Better **triage**, reduced hospital overload
 - Publishable research results (AI + Healthcare informatics)
 - Expandability to ECG, CT, Colonoscopy, Ophthalmology scans
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Impact & Atmanirbhar Bharat Alignment

- Supports **Digital Health Mission**
- Enables **affordable clinical intelligence** for India
- Reduces dependency on expensive foreign medical systems
- Scalable across **rural PHCs and clinics**

This contributes to a **self-reliant Indian healthcare ecosystem**.

Conclusion

ScanWise represents the next generation of cloud-based healthcare intelligence — combining **AI-powered multi-scan analysis**, **patient-centric communication**, and **secure triage support** for real-world healthcare improvement.

It is a **100% software solution** ready for deployment, research publication, and further enhancement as part of clinical innovation programs.
